

Phase 3: Make the agent smarter

1. Objective

The objective of this phase is to improve the Banking Support Agent by integrating a Large Language Model (LLM) into the workflow. Unlike the rule-based system developed in Phase 2, the LLM-powered agent can understand a wider range of user questions and generate more natural responses.

This phase evaluates how different prompt strategies affect response quality, safety, and consistency. Multiple prompts are tested and compared using representative banking queries. The goal is to identify a prompt strategy that provides accurate, safe, and helpful responses while minimizing hallucinations and unsafe recommendations.

2. Architecture:

Customer query → Prompt Template → LLM → Generated Response → Safe Logging

3. Python code with LLM Integration

```
import os
import re
import logging
from openai import OpenAI

# =====
# OpenAI Client
# =====

client = OpenAI(
    api_key=os.getenv("OPENAI_API_KEY"),
    base_url=os.getenv("OPENAI_BASE_URL")
)

# =====
# Configuration
# =====

# True -> Compare V1, V2, V3
# False -> Use selected default prompt only
COMPARE_MODE = True

# Selected default prompt after evaluation
DEFAULT_PROMPT_NAME = "V3"

# =====
# Logging
# =====

logging.basicConfig(
    filename="banking_agent_logs.txt",
    level=logging.INFO,
    format="%(asctime)s - %(message)s"
)

# =====
# PII Masking
# =====

def mask_pii(text):
    """
```

```

Remove potentially sensitive information before logging.
"""

# Account numbers (8+ digits)
text = re.sub(
    r"\b\d{8,}\b",
    "[ACCOUNT_MASKED]",
    text
)

# Email addresses
text = re.sub(
    r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}",
    "[EMAIL_MASKED]",
    text
)

return text

# =====
# Prompt Version 1
# =====

PROMPT_V1 = """
You are a banking support assistant.

Answer the user's banking question clearly.

User Question:
{query}
"""

# =====
# Prompt Version 2
# =====

PROMPT_V2 = """
You are a Banking Support Assistant.

Rules:
- Answer informational banking questions.
- Do not provide legal advice.
- Do not provide investment advice.
- Do not perform banking transactions.
- Escalate fraud-related complaints.
- If unsure, say you are unsure.

User Question:
{query}
"""

# =====
# Prompt Version 3 (Recommended)
# =====

PROMPT_V3 = """
You are a Banking Support Assistant.

Follow these rules:

1. Answer only informational banking questions.
2. Never invent customer information.
3. Never claim access to customer accounts.
4. Refuse requests involving money transfers.
5. Escalate fraud complaints.

```

6. If uncertain, explicitly state uncertainty.

Response Format:

Category:

Answer:

Escalation Required:

User Question:

```
{query}
```

```
.....
```

```
# Selected prompt after evaluation
```

```
DEFAULT_PROMPT = PROMPT_V3
```

```
# =====
```

```
# LLM Response Generator
```

```
# =====
```

```
def generate_response(user_query, prompt_template):
```

```
    prompt = prompt_template.format(
        query=user_query
    )
```

```
    response = client.responses.create(
        model="gpt-5.5",    # replace with model available in your environment
        input=prompt
    )
```

```
    return response.output_text
```

```
# =====
```

```
# Compare Prompt Variants
```

```
# =====
```

```
def compare_prompts(user_query):
```

```
    responses = {
        "V1": generate_response(
            user_query,
            PROMPT_V1
        ),
        "V2": generate_response(
            user_query,
            PROMPT_V2
        ),
        "V3": generate_response(
            user_query,
            PROMPT_V3
        )
    }
```

```
    return responses
```

```
# =====
```

```
# Main Agent
```

```
# =====
```

```
def main():
```

```
    print("=" * 60)
    print("Banking Support Agent (LLM Version)")
    print("=" * 60)
```

```

if COMPARE_MODE:
    print("Mode: Prompt Comparison")
else:
    print(
        f"Mode: Default Prompt ({DEFAULT_PROMPT_NAME})"
    )

print("Type 'exit' to quit")
print("=" * 60)

while True:

    user_input = input("\nCustomer: ")

    if user_input.lower() == "exit":
        print("Agent: Session Ended")
        break

    try:

        safe_input = mask_pii(user_input)

        # =====
        # Prompt Comparison Mode
        # =====
        if COMPARE_MODE:

            responses = compare_prompts(
                user_input
            )

            logging.info(
                f""""
Customer={safe_input}

PROMPT_V1:
{responses['V1']}

PROMPT_V2:
{responses['V2']}

PROMPT_V3:
{responses['V3']}

-----
""""

                )

            print("\n" + "=" * 60)
            print("PROMPT V1 RESPONSE")
            print("=" * 60)
            print(responses["V1"])

            print("\n" + "=" * 60)
            print("PROMPT V2 RESPONSE")
            print("=" * 60)
            print(responses["V2"])

            print("\n" + "=" * 60)
            print("PROMPT V3 RESPONSE")
            print("=" * 60)
            print(responses["V3"])

            # =====
            # Production Mode

```

```

# =====
else:
    response = generate_response(
        user_input,
        DEFAULT_PROMPT
    )

    logging.info(
        f""""
Customer={safe_input}
Prompt={DEFAULT_PROMPT_NAME}
Response:
{response}
""""
    )

    print(
        f"\nUsing Prompt Strategy: {DEFAULT_PROMPT_NAME}"
    )
    print("-" * 60)
    print(response)

except Exception as e:
    print(
        f"\nAgent Error: {e}"
    )

    logging.error(
        f"Error processing query: {e}"
    )

if __name__ == "__main__":
    main()

```

4. Sample Agent Interactions

```
Week 21 Capstone Project.pdf | Terminal | VS Code | voc

EXPLORER
v voc
  > course
  > coursework
  > data
  > public
  > shared
  > startercode
  > work
    .bash_history
    .nfs5207127d6...
    banking_agent...
    phase_3_make...
    Week 21 Capst...

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
Customer: What documents are required for a home loan?
=====
PROMPT V1 RESPONSE
=====
Documents typically required for a home loan include:

- **Identity proof**: Passport, driver's license, national ID, etc.
- **Address proof**: Utility bill, rental agreement, bank statement, etc.
- **Income proof**: Recent salary slips, Form 16/tax returns, or income tax returns
- **Employment details**: Employer letter or employment contract
- **Bank statements**: Usually for the last 3-6 months
- **Property documents**: Sale agreement, title deed, approved building plan, property tax receipts, etc.
- **Credit information**: Credit report or credit score authorization
- **Existing loan details**: Statements for any current loans or credit obligations
- **Photographs**: Passport-sized photos

If you're **self-employed**, you may also need business registration documents, profit & loss statements, balance sheets, and business bank statements.

Requirements can vary by bank and location, so it's best to check with your lender for the exact checklist.

=====
PROMPT V2 RESPONSE
=====
Documents required for a home loan can vary by lender, loan type, and location, but commonly include:

- **Identity proof**: Government-issued ID such as passport, driver's license, or national ID
- **Address proof**: Utility bill, lease agreement, bank statement, or other accepted proof of residence
- **Income proof**: Recent payslips, salary certificate, tax returns, or employment letter
- **Bank statements**: Typically for the last 3-6 months
- **Employment details**: Employer information and length of employment
- **Credit history/credit report**: The lender may obtain this directly
- **Property documents**: Sale agreement, title documents, property valuation, and related approvals
- **Down payment proof**: Evidence of savings or source of funds
- **Existing loan details**: Statements for any current loans or liabilities

If you're self-employed, lenders may also ask for **business financial statements, tax filings, business registration documents, and profit/loss statements**.

For the exact checklist, it's best to contact your bank or lender directly, as requirements may differ.

=====
PROMPT V3 RESPONSE
=====
Category: Home Loan Information

Answer: Documents commonly required for a home loan may include:
- Proof of identity, such as a passport, driver's license, or government-issued ID
- Proof of address, such as a utility bill or bank statement
- Proof of income, such as payslips, tax returns, or employment letters
- Bank statements, usually for the last few months
- Credit history or credit report authorization
- Details of existing debts or financial commitments
```

5. Sample Log output

The following snippet from the Vocareum shows the sample logout from the python rule based agent.

Week 21 Capstone Project.pdfTerminalVS Code

phase_3_make_agents_smarter_with_llm.pybanking_agent_logs.txt

work > banking_agent_logs.txt
1 2026-05-31 13:59:30,863 - HTTP Request: POST https://openai.vocareum.com/v1/responses "HTTP/1.1 200 OK"
2 2026-05-31 13:59:39,568 - HTTP Request: POST https://openai.vocareum.com/v1/responses "HTTP/1.1 200 OK"
3 2026-05-31 13:59:44,655 - HTTP Request: POST https://openai.vocareum.com/v1/responses "HTTP/1.1 200 OK"
4 2026-05-31 13:59:44,656 -
5 Customer=What documents are required for a home loan?
6
7 PROMPT_V1:
8 Documents typically required for a home loan include:
9
10 - **Identity proof**: Passport, driver's license, national ID, etc.
11 - **Address proof**: Utility bill, rental agreement, bank statement, etc.
12 - **Income proof**: Recent salary slips, Form 16/tax returns, or income tax returns
13 - **Employment details**: Employer letter or employment contract
14 - **Bank statements**: Usually for the last 3-6 months
15 - **Property documents**: Sale agreement, title deed, approved building plan, property tax receipts, etc.
16 - **Credit information**: Credit report or credit score authorization
17 - **Existing loan details**: Statements for any current loans or credit obligations
18 - **Photographs**: Passport-sized photos
19
20 If you're **self-employed**, you may also need business registration documents, profit & loss statements, balance sheets, and business bank statements.
21
22 Requirements can vary by bank and location, so it's best to check with your lender for the exact checklist.
23
24 PROMPT_V2:
25 Documents required for a home loan can vary by lender, loan type, and location, but commonly include:
26
27 - **Identity proof**: Government-issued ID such as passport, driver's license, or national ID
28 - **Address proof**: Utility bill, lease agreement, bank statement, or other accepted proof of residence
29 - **Income proof**: Recent payslips, salary certificate, tax returns, or employment letter
30 - **Bank statements**: Typically for the last 3-6 months
31 - **Employment details**: Employer information and length of employment
32 - **Credit history/credit report**: The lender may obtain this directly
33 - **Property documents**: Sale agreement, title documents, property valuation, and related approvals
34 - **Down payment proof**: Evidence of savings or source of funds
35 - **Existing loan details**: Statements for any current loans or liabilities
36
37 If you're self-employed, lenders may also ask for **business financial statements, tax filings, business registration documents, and profit/loss statements.
38
39 For the exact checklist, it's best to contact your bank or lender directly, as requirements may differ.
40
41 PROMPT_V3:
42 Category: Home Loan Information

6. Improvements over Phase 2

Capability	Phase 2	Phase 3
Keyword Rules	Yes	No
Natural Language Understanding	Limited	Improved
Flexible Questions	No	Yes
Banking Guidance	Limited	Better
Safety Instructions	Hardcode d	Prompt Driven
Scalability	Low	Higher

6. Baseline Agent Limitations

During testing, the selected prompt strategies frequently responded with uncertainty when information was unavailable rather than fabricating answers. Although no major hallucinations were observed in the evaluated test cases, hallucination remains a known failure mode of LLM-based systems because the model does not have access to bank policy documents or customer account data.

7. Selected Default Prompt Strategy

Selected Prompt: Version 3 (Structured Banking Prompt)

Justification

Version 3 was selected because it:

1. Produces the most consistent responses.
2. Clearly separates answer and escalation information.
3. Handles unsafe requests better.
4. Reduces hallucination risk through uncertainty instructions.
5. Supports easier evaluation and monitoring.

Although V3 uses more tokens than the other prompts, the improvement in safety, explainability, and consistency makes it the preferred prompt strategy for the Banking Support Agent.

8. Conclusion

Phase 3 successfully integrated an LLM into the Banking Support Agent workflow. Three prompt strategies were designed and evaluated using representative banking queries. The structured banking prompt (V3) provided the best balance of accuracy, safety, and consistency and was selected as the default prompt for future development. New risks such as hallucinations, overconfidence, and prompt sensitivity were identified and documented for subsequent phases.